

Name		Index No.		Civics Group	
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DUNMAN HIGH SCHOOL SENIOR HIGH

2008 PRELIMINARY EXAMINATIONS

General Certificate of Education Advanced Level

YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 01

PAPER 1 Multiple Choice

24 September 08 (Wednesday)

1 hour 15 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, civics group and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer **all** questions. For each question there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in the **soft pencil** on separate Answer Sheet [**optical answer sheet (OTAS)**] provided.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

Do not open this booklet until you are told to do so.

This question paper consists of **21** printed pages and **1** blank page (including this cover page).

Section A: Multiple Choice Questions (40 marks)

Answer all questions.

Refer to the following information from Questions 1 – 3.

To determine the intracellular pathways of three proteins X, Y and Z, four identical cultures of cells were labeled with ^3H -amino acids for 5 min.

One culture was harvested at the end of the 5 min labeling period. The other three cultures were rinsed and then were put into media with 20-fold excess of unlabelled amino acids for 10 min, 20 min and 30 min respectively before harvesting.

After harvesting, the cells were put on ice and broken open and the cellular fractions were obtained by differential centrifugation. The cellular fractions harvested included: mitochondria, rough endoplasmic reticulum (rough ER), Golgi apparatus, plasma membranes, free ribosomes and soluble cellular supernatant (cytosol).

In each of these fractions, the amount of newly synthesized protein was determined. The data, expressed as the percentage of the total newly synthesized protein of a particular type in each fraction, were recorded in the table below.

Harvest Time	5 min.			10 min.			20 min.			30 min.		
Protein	X	Y	Z	X	Y	Z	X	Y	Z	X	Y	Z
<u>Fraction</u>												
Mitochondria	0	0	0	0	0	0	0	0	50	0	0	100
Rough ER	0	100	0	0	75	0	0	10	0	0	0	0
Golgi apparatus	0	0	0	0	25	0	0	50	0	0	20	0
Plasma membrane	0	0	0	0	0	0	0	40	0	0	80	0
Ribosomes	100	0	100	25	0	0	0	0	0	0	0	0
Cytosol	0	0	0	75	0	100	100	0	50	100	0	0

- 1 The associations of proteins X and Z with free ribosomes at a harvest time of 5 min indicates which of the following about the proteins?
- A They are ribosomal proteins.
 - B They are at the site of synthesis.
 - C They are glycosylated proteins.
 - D They are identical.

[Turn over]

2 The inferred steps in the pathway of protein Y are correctly designated by which of the following?

A Endoplasmic reticulum → Golgi apparatus → plasma membrane

B Endoplasmic reticulum $\begin{matrix} \nearrow \text{plasma membrane} \\ \searrow \text{Golgi apparatus} \end{matrix}$

C Plasma membrane → Golgi apparatus → endoplasmic reticulum

D Plasma membrane $\begin{matrix} \nearrow \text{endoplasmic reticulum} \\ \searrow \text{Golgi apparatus} \end{matrix}$

3 At a harvest time of 10 min, which of the following statements is correct about protein X compared with protein Z?

A Protein X is synthesised more rapidly than protein Z.

B Protein X is degraded more rapidly than protein Z.

C Protein X is released from ribosomes more slowly than protein Z.

D Protein X is released from ribosomes more rapidly than protein Z.

4 How many fatty acid molecules are normally present in a fat or oil molecule?

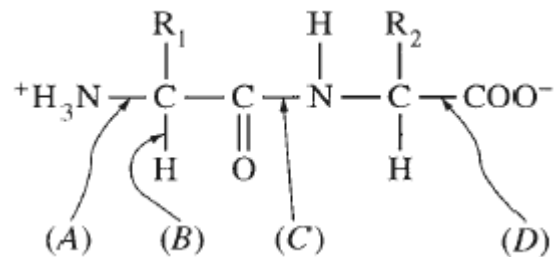
A 2

B 3

C 4

D 6

- 5 Collagen is a fibrous protein found in mammalian tendons. Which feature contributes most to the great tensile strength of collagen?
- A a quaternary structure of triple helices bonded together with covalent and hydrogen bonds
 - B a secondary structure with many hydrogen bonds firmly holding α -helices
 - C a regularly folded tertiary structure held together with hydrogen bonds and ionic bonds
 - D a primary sequence with covalent bonds linking a variable sequence of amino acids.
- 6 A peptidase hydrolyses peptide bonds in small proteins. In the molecule shown below, which bond would be hydrolyzed?



- A A
- B B
- C C
- D D

- 7 The enzyme phosphofructokinase is involved in phosphorylation of hexose phosphate sugars during glycolysis. It is involved in control of the rate of glycolysis and thus respiration, by end-product inhibition.

Deduce which of the following is a description of this enzyme.

	Shape of binding site(s)	Substrate	Product
A	no allosteric site, active site complementary to ATP and hexose	hexose	hexose phosphate
B	allosteric site complementary to glucose, active site complementary to hexose phosphate	hexose phosphate	hexose phosphate
C	allosteric site complementary to ATP, active site complementary to ATP and hexose phosphate	hexose phosphate	hexose bisphosphate
D	no allosteric site, active site complementary to hexose bisphosphate	hexose bisphosphate	two triose phosphate

- 8 In the presence of a fixed concentration of a competitive inhibitor, which of the following would best characterize an enzyme-catalyzed reaction when the concentration of the substrate is increased?
- A** the inhibition does not change
 - B** the inhibition decreases
 - C** the K_M increases
 - D** the v_{max} decreases

- 9 The table below lists the typical diploid number of chromosomes of several different organisms.

Organism	Diploid chromosome number
Goldfish	94
Potato	48
Human	46
Pea	14
Fruit fly	8

Which of the following is the **best** explanation for why the chromosome number is an even number in each of these organisms?

- A** It is only a coincidence; many other organisms have an odd number of chromosomes.
- B** The diploid chromosome number is always even so that when mitosis occurs each new cell gets the same number of chromosomes.
- C** The diploid chromosome number represents pairs of chromosomes, one from each parent, so it is always an even number.
- D** Chromosomes double every time the cell divides, so after the first division, the number is always even.
- 10 During which of the following stages of the cell cycle will a diploid cell contain twice the amount of DNA found in a gamete?
- A** prophase
- B** entire S phase
- C** entire G₁ phase
- D** entire G₂ phase

11 Which of the following describe processes that lead to an increase in variation?

- 1 Breaking and rejoining in homologous chromosomes during prophase 1 of meiosis.
- 2 Random distributions of homologous chromosomes to the cell poles during anaphase 1 of meiosis.
- 3 Random variation in allele frequency with time that may result in alleles becoming more common.
- 4 The production of new alleles by substitution of one base for another in DNA.

- A** 1 and 4 only
B 2 and 3 only
C 1, 2 and 4 only
D 1, 3 and 4 only

12 The tRNA anticodons for the amino acids arginine, proline and leucine acid, respectively are GCU, GGA and GAA.

What are the corresponding DNA triplets for these amino acids?

	arginine	proline	leucine
A	CGA	CCU	CUU
B	CGA	CCT	CTT
C	GCA	GGA	GAA
D	GCT	GGA	GAA

- 13** The diagram shows part of the normal sequence of an mRNA molecule.

CCAUGUGGUCCGUAAAAUGGC

A mutation in the DNA resulted in a polypeptide beginning with the following sequence.

Glycine – serine – proline – glycine - isoleucine

The DNA triplets for some amino acids are:

glycine	proline	leucine	serine	isoleucine
CGA	CCA	TTA	TCA	ATA
GGT	CCG	CTT	TCG	ATT
GGC		CTC		

Which mutation has occurred in the DNA molecule?

- A** an addition of an extra nucleotide
 - B** a reversal in the order of nucleotides
 - C** the loss of a nucleotide
 - D** the replacement of one nucleotide by a different nucleotide
- 14** One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4 nm long.

What is the approximate length of the DNA coding sequence of lysozyme, a protein of 129 amino acids?

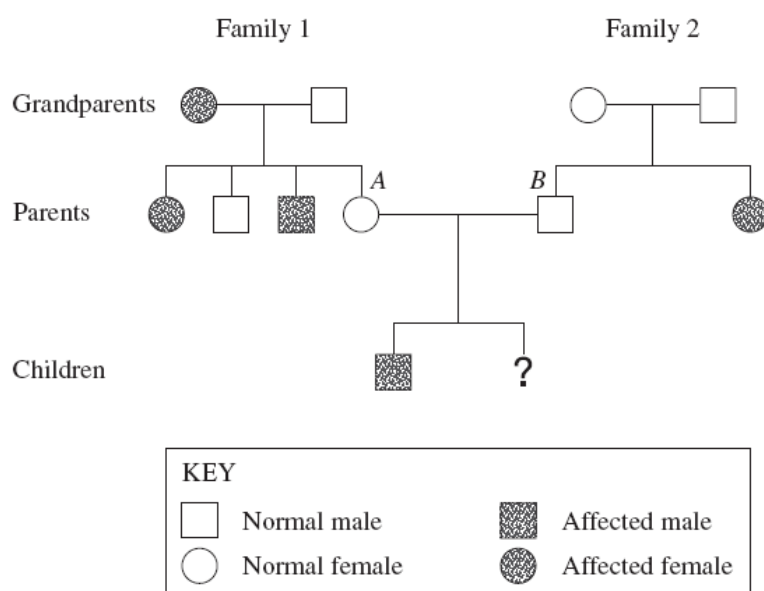
- A** 44 nm
- B** 66 nm
- C** 113 nm
- D** 132 nm

- 15 Which statement correct describes the control of transcription of the genes involved in the breakdown of lactose in *Escherichia coli*?
- A A repressor protein binds to the operator and the genes are switched on.
 - B A repressor protein binds to the operator and the genes are switched off.
 - C A transcription factor binds to the promoter and the genes are switched on.
 - D A transcription factor binds to the promoter and the genes are switched off.
- 16 All of the following are true about viruses **EXCEPT**:
- A The genome of RNA viruses are more likely to mutate than those of DNA viruses.
 - B Before entering a host cell, surface proteins of a virus bind to surface receptors of a specific host cell.
 - C All RNA viruses produce DNA as an intermediate molecule during the production of new RNA viruses.
 - D All viruses produce RNA as an intermediate molecule during the production of new DNA viruses.
- 17 A typical virus, without modification by mutation, can usually infect _____.
- A any cell in a wide variety of species
 - B any cell, but only in a single species
 - C only specific kinds of cells in a single species or in closely related species
 - D only cells with protein coats
- 18 A region of eukaryotic DNA consists of over fifty repeats of the same sequence of twelve bases. Where is this repetitive region **least** likely to be found?
- A an exon
 - B an intron
 - C a polymerase binding site
 - D satellite DNA

19 DNA fingerprints used as evidence in a murder trial look something like supermarket bar codes. The pattern of bars in a DNA fingerprint shows _____.

- A the order of bases in a particular gene
- B the presence of various-sized fragments from chopped DNA
- C the order of genes along particular chromosomes
- D the exact location of a specific gene in a genomic library

20 The family tree shows the inheritance of a genetic characteristic.



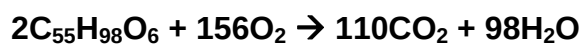
What is the probability that a daughter of parents *A* and *B* would be affected?

- A 12.5%
- B 25%
- C 50%
- D 100%

21 In carrier pigeons there is a rare inherited condition that causes the death of the chicks before hatching. In order for this disease to be passed from generation to generation there must be parent birds that _____.

- A** are heterozygous for the disease
- B** have the disease themselves
- C** produce new mutations for this disease
- D** are closely interbred

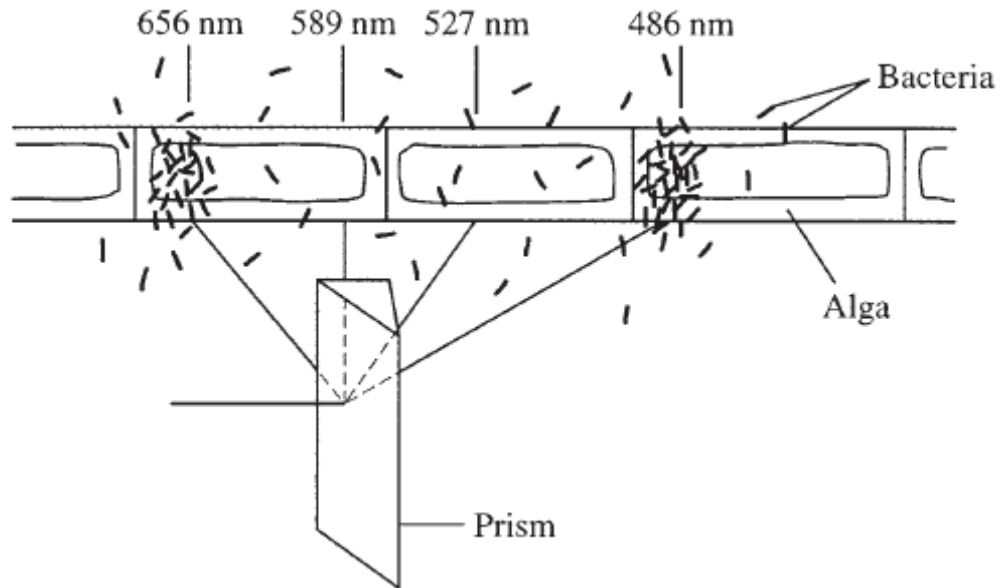
22 The equation shows the complete aerobic respiration of a respiratory substrate:



The respiratory quotient (RQ) for respiration of this respiratory substrate is

- A** 0.70
- B** 0.71
- C** 0.75
- D** 1.42

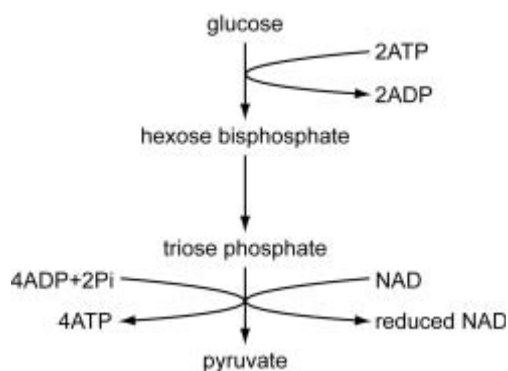
- 23** The filamentous alga *Cladophora* is illuminated with light dispersed by a prism. As shown in the diagram below, aerobic bacteria *Pseudomonas* included in the medium conglomerate where the alga is illuminated with light at 656 nm and 486 nm. *Pseudomonas* do not conglomerate in this manner if the *Cladophora* is removed from the medium.



Which of the following is the most likely explanation for the bacterial movement?

- A** Starch made by photosynthesis is secreted from the alga in regions illuminated by light at 656 nm and 486 nm.
- B** Alga and bacteria have a symbiotic relationship; the bacteria need oxygen supplied by the alga and the alga requires carbon dioxide supplied by the bacteria
- C** Oxygen evolved by photosynthesis in the regions illuminated by light at 656 nm and 486 nm attracts bacteria.
- D** Bacteria are repelled by high carbon dioxide content in regions illuminated by light at 589 nm and 527 nm.

24 The diagram summarizes glycolysis.



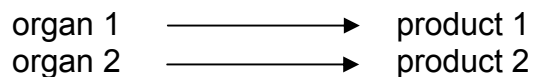
Which correctly gives the number of carbon atoms and phosphate groups in the four named molecules?

	atom/group	glucose	fructose biphosphate	glyceraldehyde phosphate	pyruvate
A	carbon	6	6	3	3
	phosphate	0	2	1	0
B	carbon	6	5	5	3
	phosphate	0	2	1	0
C	carbon	6	6	3	3
	phosphate	1	2	1	1
D	carbon	6	5	5	3
	phosphate	1	2	1	1

25 Mitochondria and chloroplasts carry out oxidative phosphorylation and photophosphorylation respectively by means of _____.

- A** photorespiration
- B** chemiosmotic coupling
- C** high-energy intermediate coupling
- D** conformational coupling

- 26** Two organs secrete substances which affect the body.



Negative feedback control of product 2 would be achieved if _____.

- A** product 1 counteracts product 2
 - B** product 1 reinforces the effect of product 2
 - C** product 2 inhibits organ 1 and product 1 stimulates organ 2
 - D** product 2 stimulates organ 1 and product 1 stimulates organ 2
- 27** When a stimulus arrives at a nerve ending, the membrane becomes depolarized. The events during depolarization are listed below.

- 1 K⁺ channels open, Na⁺ channels close.
- 2 Na⁺ channels open and Na⁺ floods in
- 3 Potential difference inside the membrane is +40 mV compared to the outside.
- 4 Potential difference inside the membrane is -65 mV compared to the outside.

What is the correct order of events?

- A** 1, 2, 3, 4
- B** 2, 4, 1, 3
- C** 3, 4, 1, 2
- D** 4, 2, 3, 1

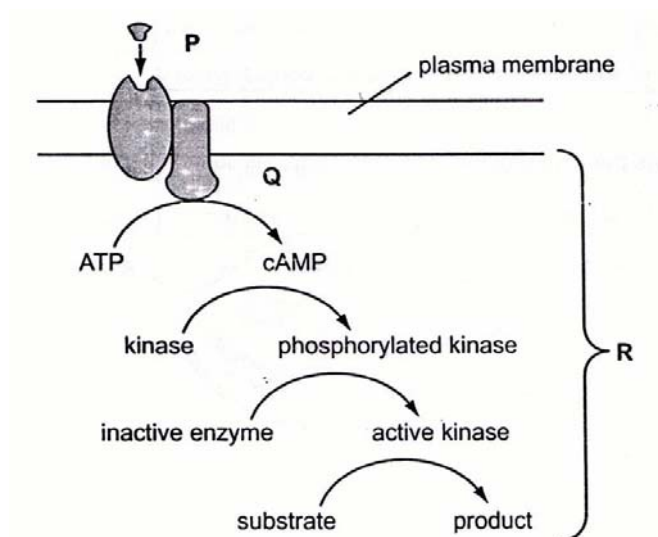
28 Which is a correct description of the role of calcium ions in the neuromuscular system?

- A** exchanged with sodium ions through co-transport channels at axon surfaces during the reestablishment of a resting potential after an action potential
- B** moved in by diffusion through gated ion channels in pre-synaptic membranes of excitatory neurones causing vesicles to move to pre-synaptic membrane as an impulse arrives
- C** exchanged with chloride ions at the post-synaptic membrane, in changing membrane potential in inhibitory neurones
- D** actively pumped out of axons at nodes of Ranvier of myelinated neurones, being the main cause of the potential difference that is maintained during the resting potential

29 A person has poorly functioning β cells in the islets of Langerhans. What will be the concentration of glucose, insulin and glucagon in the blood after that person has eaten a meal rich in carbohydrates?

	glucose	insulin	glucagon
A	high	low	high
B	high	low	low
C	low	high	high
D	low	high	low

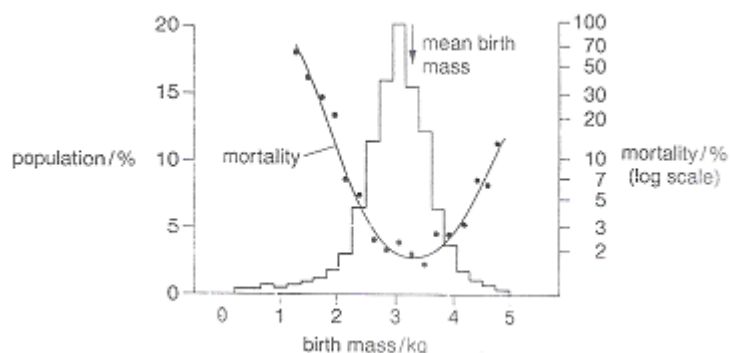
30 The diagram shows an example of cell signaling.



Which processes are shown on the diagram?

	P	Q	R
A	amplification	ligand-receptor interaction	phosphorylation
B	ligand-receptor interaction	amplification	phosphorylation
C	transduction	phosphorylation	amplification
D	ligand-receptor interaction	transduction	amplification

31 Various factors have influenced the evolution of humans.

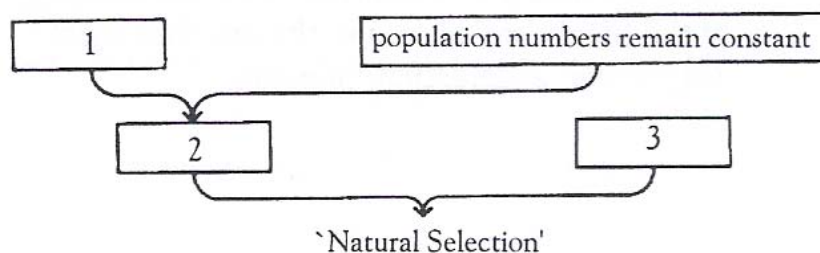


The graph above illustrates one factor.

Which factor does the graph illustrate?

- A Artificial selection
 - B Directional selection
 - C Disruptive selection
 - D Stabilising selection
- 32 A large lake evaporates slowly over time, leaving many small ponds. The large lake had been inhabited by one species of fish, but over time, each small pond is inhabited by a different species of fish, each species a descendant of the species which inhabited the large lake. This is an example of _____.
- A punctuated equilibrium
 - B disruptive selection
 - C allopatric speciation
 - D sympatric speciation
- 33 What prevents horses and donkeys from hybridising to form new species?
- A hybrid sterility
 - B hybrid inviability
 - C hybrid breakdown
 - D prezygotic barrier

- 34 The diagram shows a summary of Darwin's theory of natural selection.



Which statements should be placed in boxes 1, 2 and 3?

	Individuals show great reproductive capacity	Variation is shown in all populations	There is a struggle for existence
A	1	2	3
B	1	3	2
C	2	3	1
D	3	1	2

- 35 A researcher wants to clone a human gene whose primary RNA transcript contains introns. A bacterial plasmid is used as a vector to transfer the gene into bacteria. To ensure success of the procedure, all of the following steps should be included in the process **EXCEPT**:
- A** Treat the plasmid and the human DNA with the same restriction enzyme.
 - B** Use a restriction enzyme that does not produce sticky ends.
 - C** Use the complementary DNA (cDNA) of the human gene.
 - D** Provide suitable conditions (such as a calcium-rich growth medium) for transformation of the plasmid by the bacteria.

36 Five stages in the production of human insulin by genetic engineering techniques are given.

- 1 DNA cut with restriction enzymes
- 2 DNA copy made using reverse transcriptase
- 3 Messenger RNA extracted from cells
- 4 Plasmid DNA joined to donor DNA using ligase enzymes
- 5 Recombinant plasmid inserted into bacterial cell

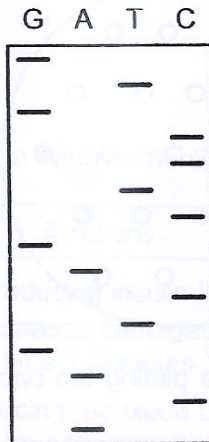
Which sequence is correct?

first $\longrightarrow$ last

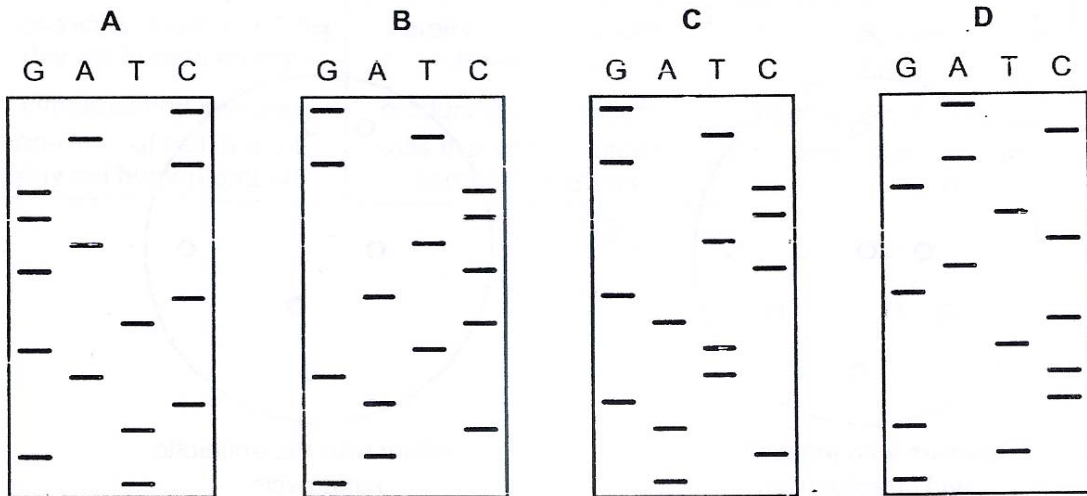
- A** 1 $\rightarrow$ 3 $\rightarrow$ 2 $\rightarrow$ 5 $\rightarrow$ 4
B 1 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 2 $\rightarrow$ 5
C 2 $\rightarrow$ 1 $\rightarrow$ 3 $\rightarrow$ 5 $\rightarrow$ 4
D 3 $\rightarrow$ 2 $\rightarrow$ 1 $\rightarrow$ 4 $\rightarrow$ 5

- 37 In people with a mutant allele, a protein contains just one different amino acid in its primary structure. To identify the presence of this mutant allele, DNA nucleotide sequences were compared using electrophoresis.

The electrophoresis results from the DNA of a normal allele of a gene is shown below.



Which diagram represents the DNA sequence for the mutant allele of this gene?



- 38 Which type of gene therapy is **least** effective in transferring DNA to the nucleus?

- A adenoviruses
- B DNA attached to a ligand
- C liposomes
- D retroviruses

39 What are the normal functions of stem cells in a living human?

	Functions		
A	differentiating into many kinds of cells in 3 – 5 day old human embryo	Producing insulin in pancreases damaged by type 1 diabetes	Cells that can differentiate into bone cells in the skeleton
B	differentiating into many kinds of cells in 3 – 5 day old human embryo	Producing red blood cells worn out by normal wear and tear	Cells that can differentiate into cardiac muscle cells in the heart
C	differentiating into many kinds of cells in 3 – 5 day old human embryo	Producing dopamine in the brains of people with Parkinson's	Cells that can differentiate into cartilage cells in the joints
D	differentiating into many kinds of cells in 3 – 5 day old human embryo	Producing differentiated cells that can be used to screen new drugs	Cells that can differentiate into nerve cells in the brain

40 Synthesis of human insulin by genetically manipulated bacteria involves the use of the enzyme reverse transcriptase.

What is the role of this enzyme?

- A** It adds 'sticky ends' to insulin genes.
- B** It causes production of mRNA in the pancreas cells.
- C** It causes complementary DNA to be formed from mRNA.
- D** It causes single-stranded DNA to convert to double-stranded DNA.

END OF PAPER

Name		Index No.		Civics Group	
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DUNMAN HIGH SCHOOL SENIOR HIGH

2008 PRELIMINARY EXAMINATIONS

General Certificate of Education Advanced Level

YEAR SIX H2 BIOLOGY (9747)

BIOLOGY

9747 / 02

PAPER 2 Core Paper

12 September 08 (Friday)

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST.

Write your name, civics group and index number on all the work you hand in.

Write in dark blue or black pen.

You may use a soft pencil for any diagrams, graphs or rough working.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Do not open this booklet until you are told to do so.

Section B:

Consists of 9 Structured Questions

Answer **all** questions.

Write your answers in the **space provided** on the question paper.

Section C:

Consists of 2 Essay Questions

Answer any **one** question.

Write your answers on the separate **writing papers** provided.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE		
Section A (MCQ)		
Section B	1	
	2	
	3	
	4	
	5	
	6	
	7	
	8	
	9	
Section C	10 / 11	
Total		

This question paper consists of **23** printed pages and **1** blank page (including this cover page).

Section B: Structured Questions (80 marks)

Answer **all** questions in this section.

Question 1

- (a) Beer contains ethanol. Suggest and explain the role of yeast enzymes in beer production. [3]

- (b) Most beers contain starch. Recently, 'light' beers of low energy content have become more popular. Light beers have low starch content. This is achieved by the addition of immobilized fungal amylase after the mashing process.

- (i) Suggest two advantages of using immobilized enzymes in this process. [3]

The effect of two different types of immobilized fungal amylase on the hydrolysis of starch is shown in **Table 1**. In these reactions, starch is not a limiting factor.

Table 1

Time/hr	Mass of maltose produced / g	
	α amylase	β amylase
0	0	0
1	0.05	0.05
2	0.20	0.10
3	0.60	0.20

- (ii) **State** two other factors not mentioned that the beer producers have to adjust in order to maximise their production efficiency. Briefly **explain** why this is so. [2]

- (iii) With reference to **Table 1**, explain which of these enzymes would be used in the production of light beers with low starch content. [2]

Total: [10]

Question 2

Table 2 shows the nucleotide sequences for some of the amino acids.

Table 2

CUU* leucine	CCU Proline	CAA glutamine	CGU arginine
AUU isoleucine	ACU threonine	AAU asparagine	GUU valine
GCU alanine	GAA glutamic acid	GGU glycine	UGG tryptophan

*messenger RNA codon corresponding to the amino acid

- (a) Using the information provided in **Table 2**, write out the base sequences that would produce the following chain of amino acids for

glycine – isoleucine – valine – glutamic acid – glutamine
in terms of

- (i) messenger RNA (mRNA). [1]

- (ii) DNA. [2]

- (b)** The amino acids are assembled with the help of tRNA. Describe the structural features of a tRNA molecule which help it to adapt to its role in translation. [2]

Total: [5]

Question 3

Bacteria have three means of transferring genetic information.

(a) Name and briefly explain the three recombination mechanisms. [6]

- (b)** In a laboratory, a student starts with two strains of bacteria:
- Strain A: resistant to penicillin, not resistant to ampicillin
 - Strain B: not resistant to penicillin, resistant to ampicillin

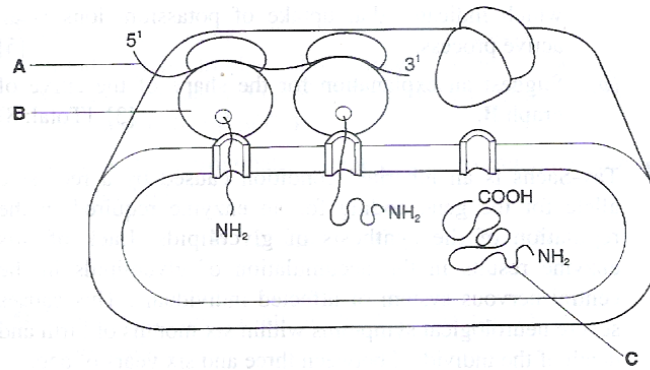
Design a procedure by which the genes that code for penicillin resistance are transferred to the strain that is not penicillin resistant; the resulting strain would therefore be resistant to both penicillin and ampicillin. Identify the recombination mechanism used. [2]

- (c) How would the student isolate the strain which is resistant to both ampicillin and penicillin? [2]

Total: [10]

Question 4

Figure 4 is a diagram of a section through the rough endoplasmic reticulum representing the synthesis of proteins.

**Figure 4**

(a) Name structures **A** to **C**. [3]

(b) Describe the role in protein synthesis of

(i) Structure **A**; [1]

(ii) Structure **B**. [1]

(c) Explain the meaning of the term cancer. [2]

(d) Briefly describe how cancer arises. [3]

Total: [10]

Question 5

Different inbred lines of chickens show pronounced differences in mortality following infection with equal doses of two different strains of the bacterium *Salmonella typhimurium*.

In an investigation into the inheritance of resistance, a series of crosses was performed between two inbred lines of White Leghorn chickens, one line resistant to infection by the bacterium and one susceptible. The results of infecting newly hatched chicks of each inbred line, and of the cross between a susceptible male and a resistant female, are shown in **Table 5.1**.

Table 5.1

line of chickens	mortality / % of chickens infected	
	strain of <i>S. typhimurium</i>	
	F98	Swindon
resistant	40.0	0.0
susceptible	100.0	35.7
cross between susceptible ♀ and resistant ♂ (F ₁ hybrid)	45.0	0.0

It is thought that resistance to *S. typhimurium* is controlled by a single gene.

- (a) (i) Using suitable symbols, draw a genetic cross to show how resistance to *S. typhimurium* could be inherited in the cross that produces the F₁ hybrid shown at the bottom of **Table 5.1**. [3]

Working for genetic cross

Working for genetic cross (cont'd)

- (ii) Predict the ratio of resistant to susceptible chicks resulting from crosses between the following birds: [3]

Susceptible x F₁ hybrid

Resistant x F₁ hybrid

F₁ hybrid x F₁ hybrid

[Turn over]

When 20 chicks from crossing a susceptible male and a resistant female were infected with equal doses of strain F98 of *S. typhimurium*, 9 died, but when 19 chicks from the reciprocal cross of a resistant male and a susceptible female were infected with equal doses of the same strain of bacterium, only 3 died. In both cases, 8 chicks (40%) were expected to die.

- (b) (i) State **two** reasons why the observed results of such crosses may be different from the expected results. [2]

- (ii) Using the chi-squared test and the table of probabilities shown in **Table 5.2**, calculate the probability of the results of the reciprocal cross (resistant male and susceptible female) departing significantly by chance from expectation. Show your working and conclusion clearly in the space below. [5]

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

$$v = c - 1$$

Where Σ = 'sum of ...'
 v = degrees of freedom
 c = number of classes
 O = observed 'value'
 E = expected 'value'

Table 5.2

degrees of freedom	probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

Working for chi-squared test

Total: [13]

[Turn over]

Question 6

2,4-dinitrophenol is a chemical that is toxic to mitochondria. When added to mitochondria this chemical allows electron transport to occur but prevents the phosphorylation of ADP to ATP. The chemical achieves this by breaking the essential link between electron transport and ATP synthesis. This toxin causes mitochondria to produce heat instead of ATP. The greater the amount of toxin added, the quicker is its action.

- (a) If mitochondria are poisoned with 2,4-dinitrophenol by what process could a plant cell produce more ATP? [1]

- (b) A researcher wanted to study cellular respiration in insect cells. She cultured some muscle cells from the common field cricket, *Teleogryllus oceanicus*, and then studied the effects of adding 2,4-dinitrophenol to these cells. An agricultural company may want to fund this research.

Suggest why an agricultural company might want to fund research on the effects of this toxin on field crickets. [1]

The experiment is summarized in **Table 6** below. Temperature observations in each trial were made at equal time intervals.

Table 6

Observations made at equal time intervals	Temperature / °C		
	Control (no 2,4- dinitrophenol)	Trial 1 (2,4- dinitrophenol added)	Trial 2
1 (start)	28	28	28
2	27	28	29
3	28	29	
4	29	31	
5	28	36	
6	28	23	
7	27	21	

- (c) Explain why the temperature went down after the fifth observation in trial 1. [1]

- (d) Trial 2 had twice the concentration of 2,4-dinitrophenol added.

Complete the table by writing in temperatures in the spaces provided to predict the trend. [2]

Another researcher suggested adding pyruvate to the cells at the beginning of the experiment to cancel out the effects of this toxin.

- (e) Explain what effect adding pyruvate would have on cancelling out the effect of this toxin. [2]

Total: [7]

Question 7

- (a) The bird's nest fern, *Alocasia macrorrhiza*, usually grows in deeply shaded rainforests and has dark green fronds (leaves). Sometimes it is found in open, sunny locations by roadsides where it tends to have lighter colored fronds.

Two bird's nest ferns, one from each of the two habitats described above, were examined. A sample of cells from a frond of each of the ferns was collected. These cells were examined under an electron microscope and a typical chloroplast from each habitat (deeply shaded rainforest and sunny location) was drawn. These drawings are shown below in **Figure 7.1**.

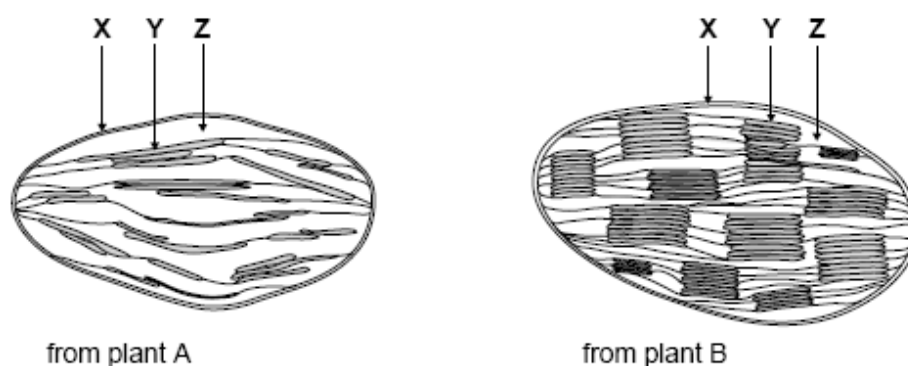


Figure 7.1

- (i) Which of the labeled parts, X, Y or Z, absorb light energy? **Name** the part. [1]

- (ii) Which drawing, **A** or **B**, shows a chloroplast from the rainforest habitat? Explain the reason for your choice in terms of the relationship between structure and function. [2]

- (b)** Chloroplasts contain a large number of ribosomes.

Explain the importance of the presence of these ribosomes. [2]

- (c)** It is now widely accepted by biologists that chloroplasts and mitochondria were once independent prokaryotic organisms which came to live symbiotically inside larger eukaryotic cells. This idea is known as the endosymbiotic theory.

Name two structural features of chloroplasts or mitochondria and outline how they support the endosymbiotic theory for the origin of these organelles. [2]

- (d) The simplified diagram **Figure 7.2** below summarizes the light-independent reactions (Calvin cycle) of photosynthesis.

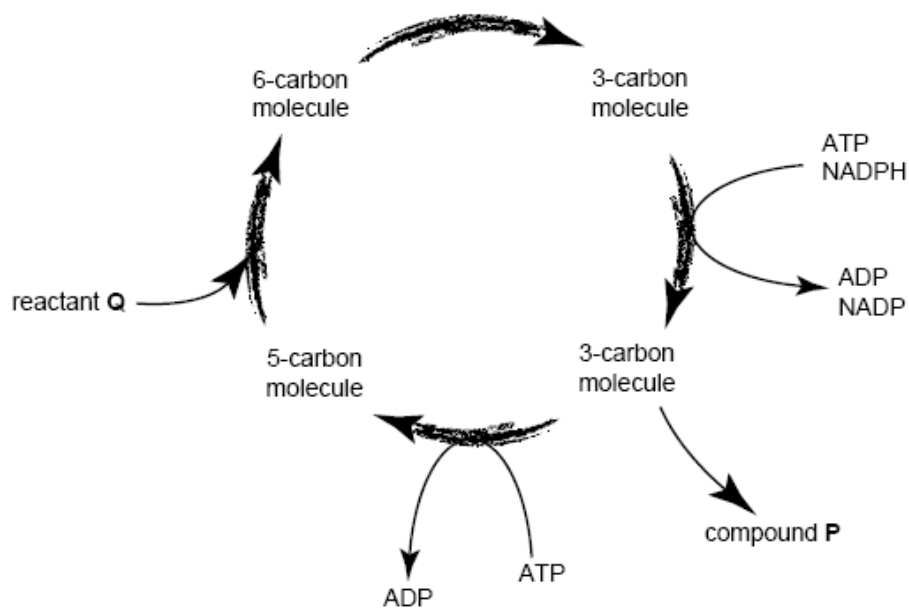


Figure 7.2

Name compound **P** and reactant **Q**. [2]

P:

Q:

Total: [9]

Question 8

Cell A and cell B represent two types of cells found in complex multicellular animals. Cell A is a nerve cell (neurone). Cell B is an endocrine cell. **Figure 8.1** below shows how cells A and B interact with other types of cells, cell X and cell Y, in the body.

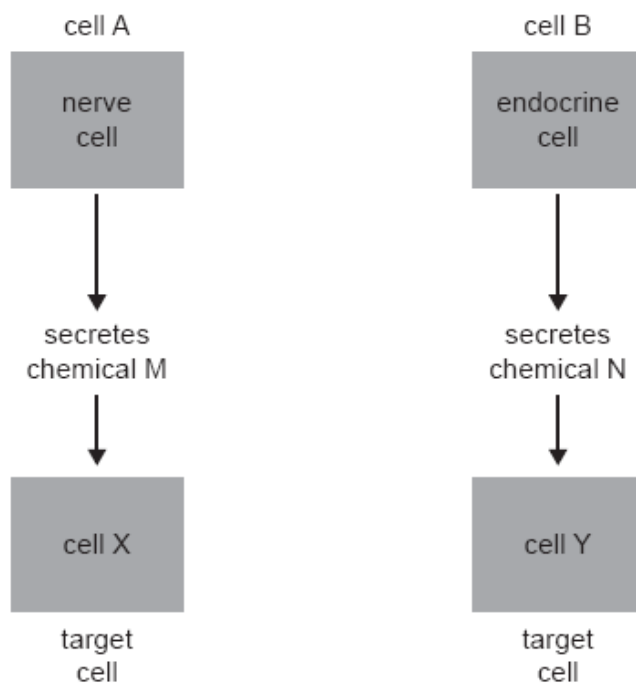


Figure 8.1

(a) Name chemical **M** and describe how it reaches cell **X**. [2]

(b) Name chemical **N** and describe how it reaches cell **Y**. [2]

- (c) Does the nervous system or the endocrine system have a quicker response? Explain why this is so. [2]

- (d) A tissue may, over time, lose its response to a particular hormone, even though the hormone concentration remains unchanged.

Based on your understanding of how a hormone controls the functioning of other cells, suggest a reason for this decrease in responsiveness. [1]

- (e) The concentration of glucose in human blood is kept within a narrow normal range. All body cells need a continuous supply of glucose. To prevent the concentration of glucose in the blood falling below the normal range the pancreas releases a hormone.

Explain how this hormone causes an increase in the blood glucose concentration. [2]

- (f) When the concentration of glucose in the blood rises above the normal range the pancreas releases another hormone. Some humans cannot make enough of this hormone and must have regular injections of this hormone.

Different forms of this hormone have been produced by pharmaceutical companies.

Table 8.2 below shows information on three forms of the hormone.

Table 8.2

Different forms of hormone	Onset of action	Peak action	Duration
A	15 min	30-90 min	3-5 hrs
B	30 min	2 – 4 hrs	6 – 8 hrs
C	4 – 8 hrs	12 – 18 hrs	24 – 28 hrs

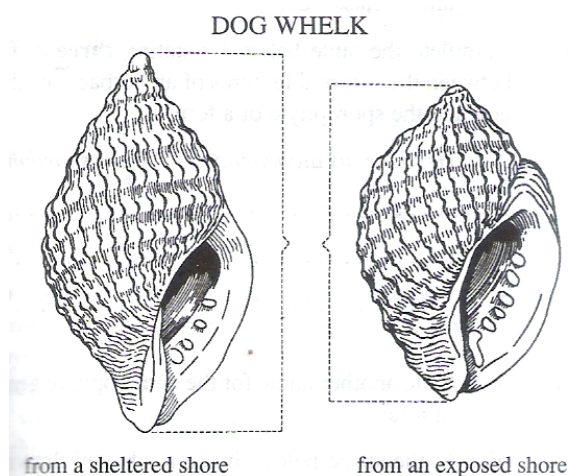
- (i) State one advantage of form C. [1]

- (ii) Describe one situation when it would be advantageous to use form A. [1]

Total: [11]

Question 9

The dog whelk (*Nucella lapillus*) is commonly found on rocky sea shores around the coasts of the British Isles. Considerable variability of its shell thickness and shape occurs on shores exposed to different strengths of wave action, as shown in **Figure 9**. On sheltered shores they have a high, thick shell and a relatively small aperture for attachment to rocks; on exposed shores the shell is thinner and stubbier, so that they are better able to hold on to the rocks they inhabit when subjected to the pounding of waves. The main predators of dog whelks are crabs, which will be active almost entirely on sheltered shores. Shell thickness is clearly protective here.

**Figure 9**

Describe the role of natural selection, in maintaining the shell shape and thickness of populations of dog whelks on:

(a) (i) Sheltered rocky shores, [2]

(ii) Exposed rocky shores. [3]

Total: [5]

[Turn over]

Section C: Free-response Questions (20 marks)

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, wherever appropriate.

Your answer must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 10 (a)** Describe the structure of the cell membrane. Relate its structural components to its functions. [14]
- (b)** The phospholipid bilayer forms an essential part of living cells but can also serve as a hindrance to certain cellular processes. Highlight two distinct examples and explain how the cell copes with this problem. [6]
- 11 (a)** Explain Darwin's theory of natural selection. [6]
- (b)** Describe two pieces of evidences which suggest that natural selection is still occurring. [8]
- (c)** What part does natural selection play in the evolution of new species? [6]

END OF PAPER